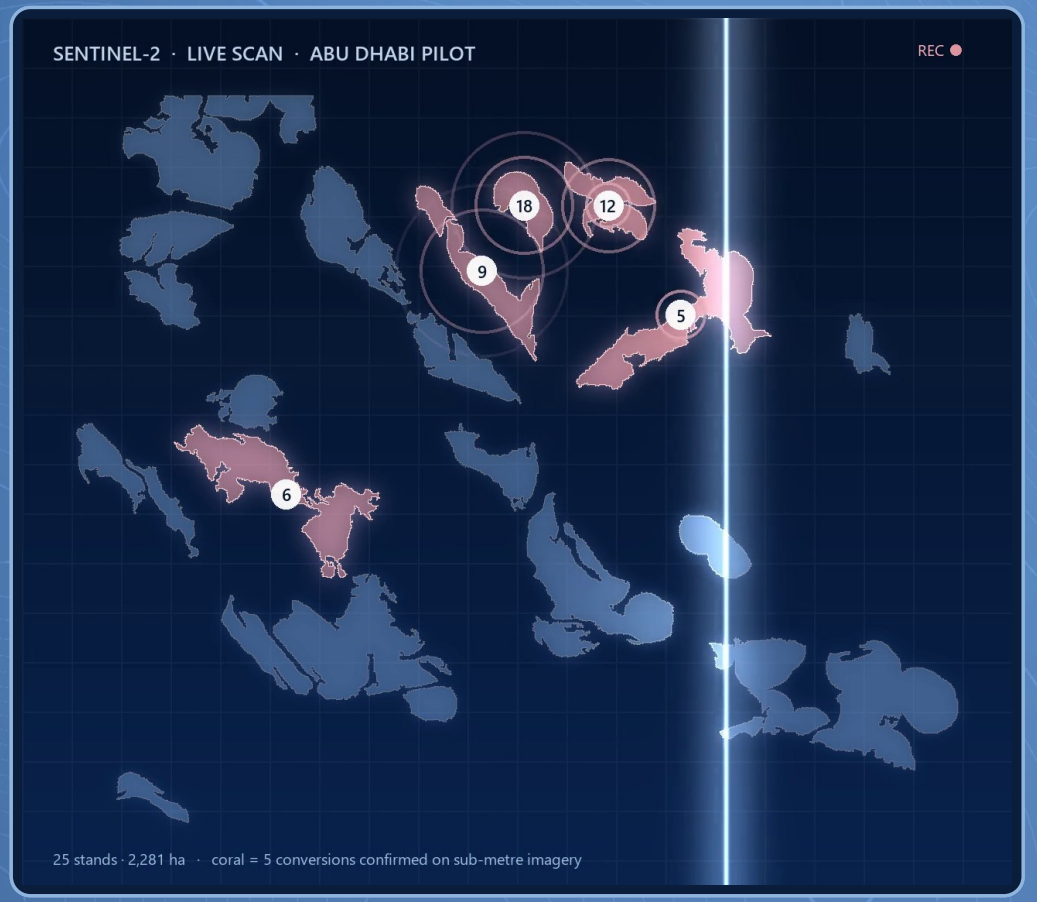
ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · FINAL

# Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves, from alert to evidence.


Team: Blue Athar · الأثر الأزرق


Theme: Ecosystem Health, Biodiversity & Blue Carbon




| 91 | 5 | 73 % | ≈ 1 |
|---|---|---|---|
| stands monitored | conversions confirmed | partial loss caught (chance 13 %) | false alarm per stand-year |

Study area: Abu Dhabi, UAE

Built in Bahrain


**Nahla Nabil · Bahrain**

THE TEAM

# A solo project: one engineer, every role



## Nahla Nabil

Solo founder · AI engineer

Bahrain

linkedin.com/in/nahla-nabil-876597211

### EDUCATION

**BSc Artificial Intelligence** · Arab Open University, Bahrain, UK · final semester · First-Class Honours standing

### RESEARCH

- Current focus: AI for measuring blue carbon
- 2 IEEE papers: ICCA 2025 and ICAISET 2026
- **3rd paper, presenting 29 Oct 2026: PIGNN-UQ, uncertainty-aware blue carbon crediting**

### EXPERIENCE

- IT intern · Ministry of Interior, Bahrain
- AI & LLM engineering · Doo Bahrain, FlyRank US
- Data Analysis

### COMMUNITY

**NASA Space Apps Challenge Local Lead, Manama 2026**

### ROLES & RESPONSIBILITIES IN THIS PROJECT

**Project lead & owner**

problem framing, user needs, scope, every claim

**EO data engineer**

778 Sentinel-2 dates, EnMAP reader, sub-metre checks

**Modelling & statistics**

tide-aware model, stand + cell alert, calibration

**Validation & integrity**

real events, simulations, unseen stands

**Product & delivery**

dashboard, website, site reports, notebook

**Business & impact**

customers, pricing hypotheses, carbon ranges

**Incubation gap we will close:** a mangrove ecologist mentor · a restoration or agency partner · a GIS analyst

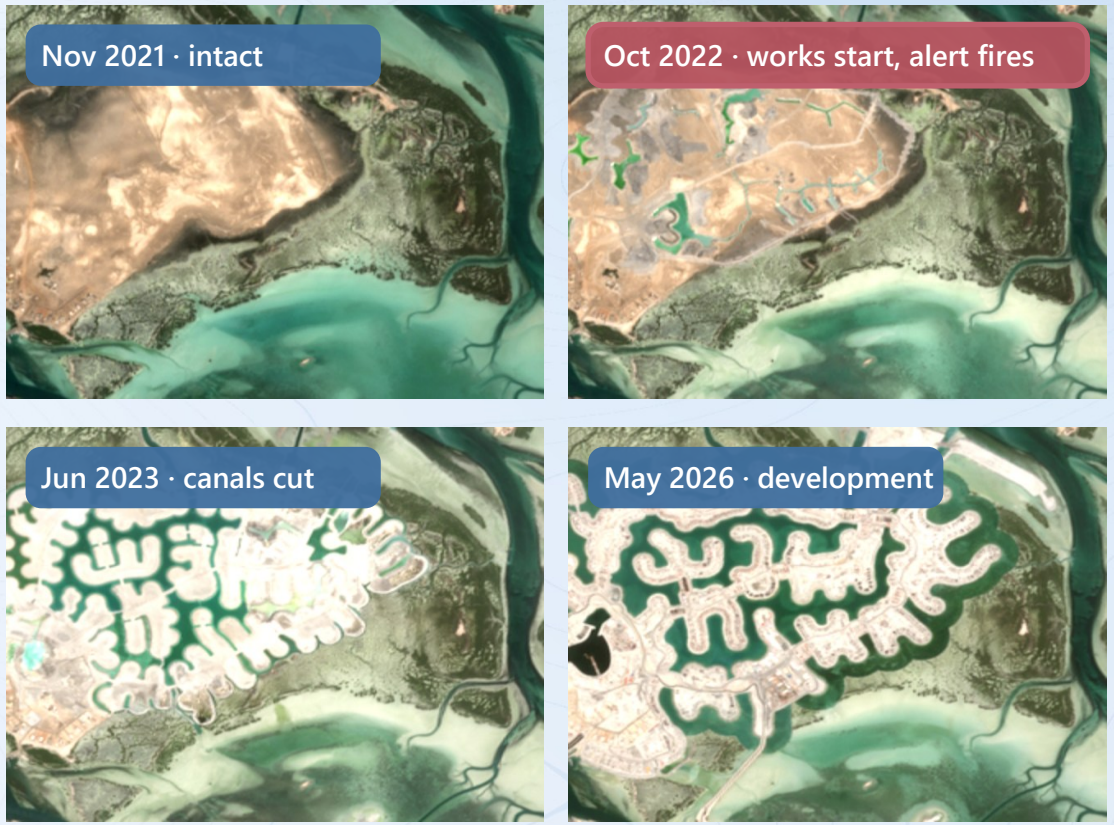
THE PROBLEM

# Restoration is scaling faster than monitoring

| | |
|---|---|
| 100 M | **UAE mangrove pledge by 2030**<br>COP26, 2021 · Abu Dhabi Mangrove Initiative |
| 100 M+ | **Saudi mangrove pledge by 2030**<br>Saudi Green Initiative |
| ×4 | **Bahrain mangrove pledge by 2035**<br>COP26, 2021 · national net-zero path |
| Feb 2026 | **first GCC mangrove monitoring guide**<br>EAD + IUCN Mangrove Specialist Group |

**TODAY** field surveys and one-off studies, no standing alert.
**WE SOLVE** a calibrated alert every ~5 days that says where to inspect and what carbon is at stake.

THIS IS HAPPENING NOW · STAND 5, ABU DHABI




Sentinel-2 true colour (contains modified Copernicus Sentinel data). Our alert began on 25 Oct 2022, during the works.



Sources: UAE 100 M by 2030 — MOCCAE, COP26 (2021) · Saudi 100 M+ — Saudi Green Initiative · Bahrain ×4 by 2035 — COP26 (2021) · GCC mangrove monitoring guide — Environment Agency – Abu Dhabi with IUCN Mangrove Specialist Group (Feb 2026) · Guidelines for Mangrove Restoration in the UAE — EAD, MOCCAE, Emirates Nature–WWF

LIVE PRODUCT · ABU DHABI PILOT

# The monitor

25 mangrove stands (2,281 ha), Sentinel-2 from 2020 to October 2026. Click a stand on the map or in the table. Switch to **200 m cells** to see where inside a stand the change is.

Stands | 200 m cells

Colour = condition, last 12 months vs 2020–21



1 · Every stand coloured by its condition

## Stand 5

severe decline

No active alert. 3 alert episodes 2022–26 (combined stand + cell rule).

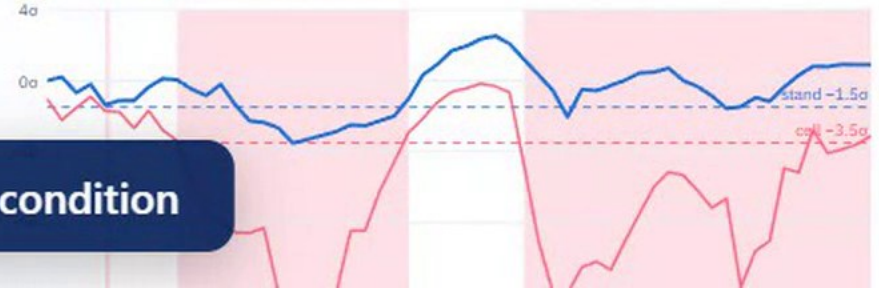
**176.5 ha**
stand area

**19.1 kt C**
carbon stock (13.9–24.4)

**-111 %**
NDVI, last 12 months vs 2020–21

**27 of 45 cells**
in decline · 72.2 ha to inspect

ALERT SCORE, 2022–2026



BUSINESS USE CASE & END USERS

# Who uses it, and for which decision

**Growth and protection, together:** the UAE is planting 100 million mangroves as it builds its coast. This monitor helps both succeed.

## Environment agencies

EAD, MOCCAE, Saudi NCVC

THEIR DECISION

**Which stands do we inspect this month?**

**Today:** periodic field surveys

monthly alert + 200 m cell map: inspect only flagged cells

## Planting & carbon developers

ADNOC, Red Sea Global, carbon projects

THEIR DECISION

**Is the carbon we report still there?**

**Today:** plots and drone campaigns

condition + carbon stock with a range, a site report

## Developers & regulators

EIA consultants, permit offices

THEIR DECISION

**Did works stay outside the mangrove line?**

**Today:** comparing two images by eye

an alert when works enter mangrove cells, dated evidence

**New scene** every ~5 days → **Alert** ~1 false alarm / stand-yr → **Cell map** where to look → **Inspection** only flagged cells → **Site report** carbon at stake

# Five independent data sources, six years, 91 stands




Pilot: 25 stands · 2,281 ha (WorldCover 2021 patches ≥ 3 ha). +66 unseen stands in Abu Dhabi N/W and Saudi Tarut Bay test transfer.

### Sentinel-2 L2A
ESA / Copernicus · 10–20 m time series
778 dates

### EnMAP hyperspectral
DLR · 224 bands · independent check
2 scenes

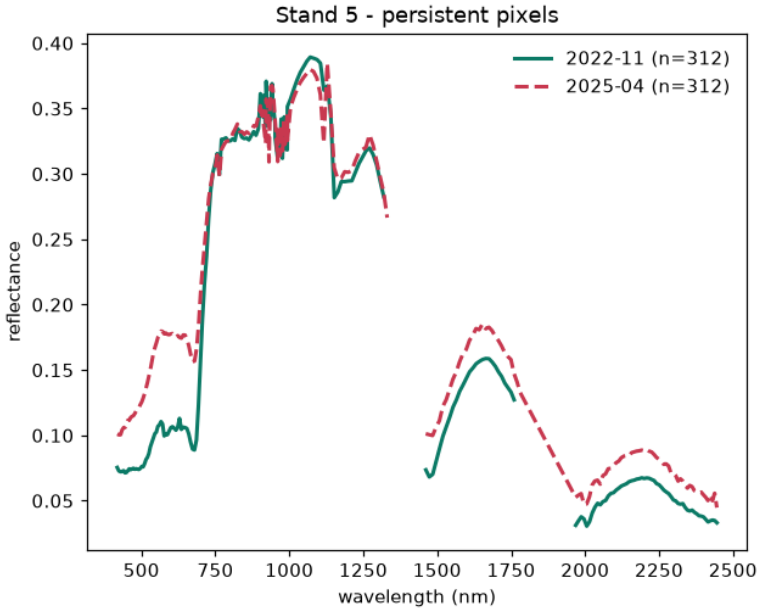
### ESA WorldCover 2021
10 m land cover · stand outlines
CC BY 4.0

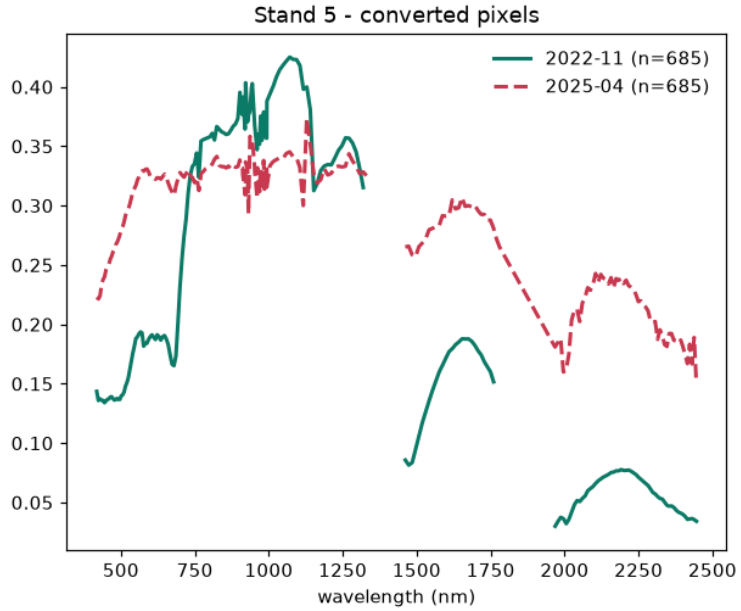
### Field carbon plots
Schile et al. 2016 · carbon density
24 plots

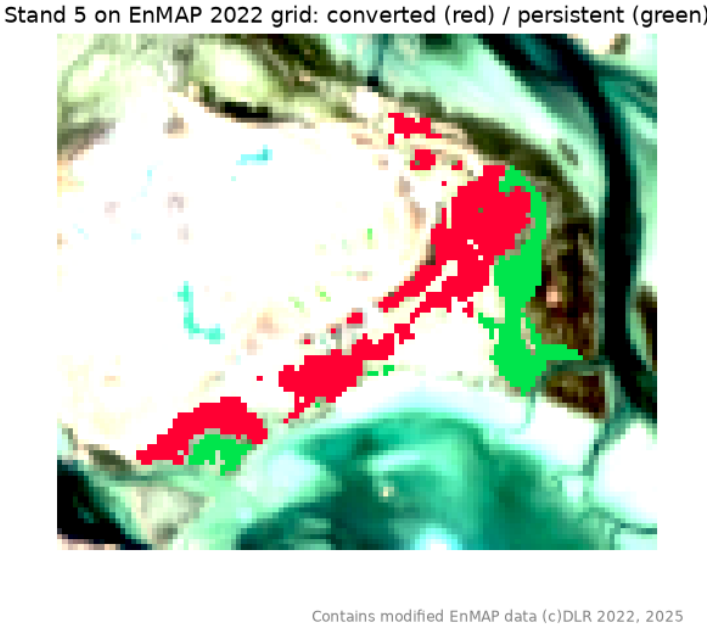
### Sub-metre captures
WorldView-2/3, Legion-1 · confirm events
0.3–0.5 m

Quality control: cloud masks · interior pixels only · tide model · second-source checks

# A second sensor shows what the mangrove became




## r = 0.84–0.99

agreement with our Sentinel-2 indices on the same dates (25 and 14 stands)

## 62 vs 26 ha

stand-5 loss: EnMAP pixel count vs our conservative Sentinel-2 flag

## Vegetation → mineral

converted pixels take the spectrum of sand and fill: reclamation, not dieback

Honest limit: as a classifier, the full spectrum is no better than Sentinel-2-like bands (F1 0.74 vs 0.72). Contains modified EnMAP data © DLR 2022, 2025.

# Sentinel-2 says when. Satellite 813 says what.

| Sensor | Bands | Range | Pixel | Revisit | Role in the product |
|---|---|---|---|---|---|
| **Sentinel-2** | 13 | 443–2190 nm | 10–20 m | ~5 days | WHEN: the alert, every pass |
| **EnMAP (used)** | 224 | 420–2450 nm | 30 m | ~27 days | WHAT: tested on stand 5 |
| **Satellite 813 (next)** | **~205** | **~400–1700 nm** | **~20 m** | **on demand** | **WHAT, sharper: tasked on flagged cells** |

### 1 · Sentinel-2 alert

stand or 200 m cell below its threshold

→

### 2 · Task Satellite 813

only over the flagged cells, on demand

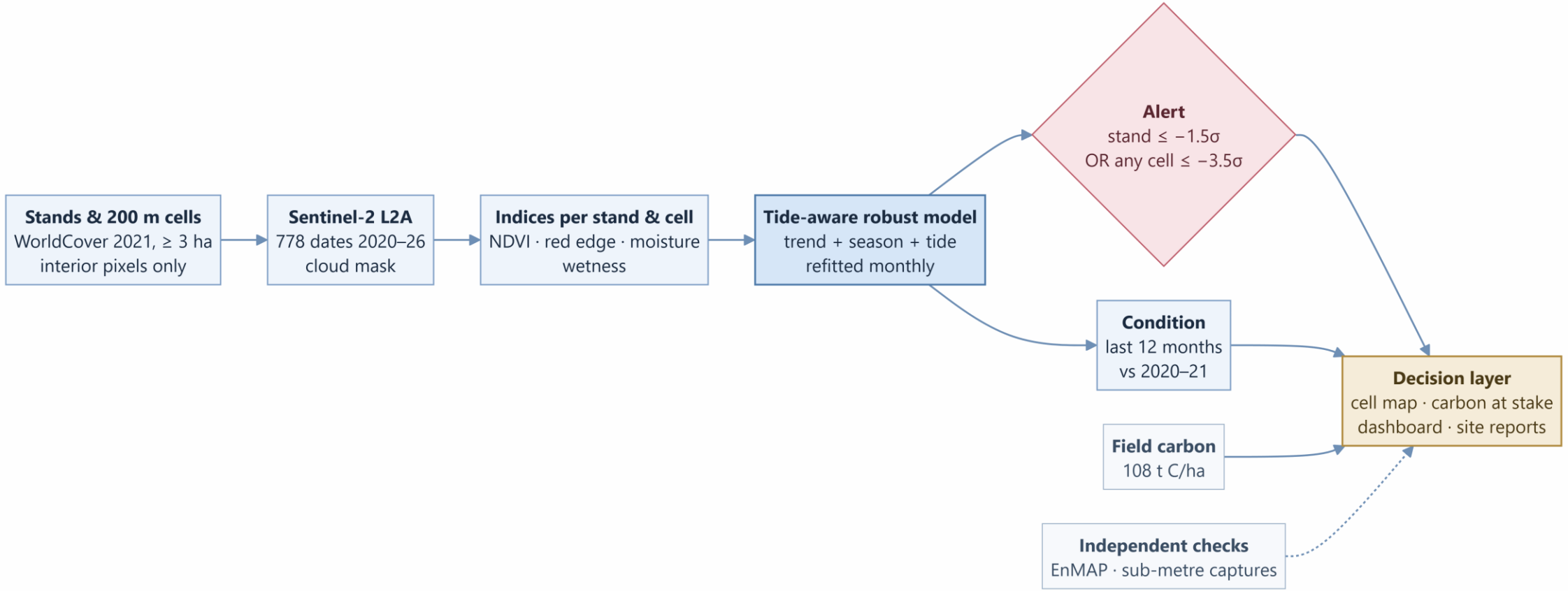
→

### 3 · Spectral diagnosis

vegetation, or sand and fill: an evidence pack

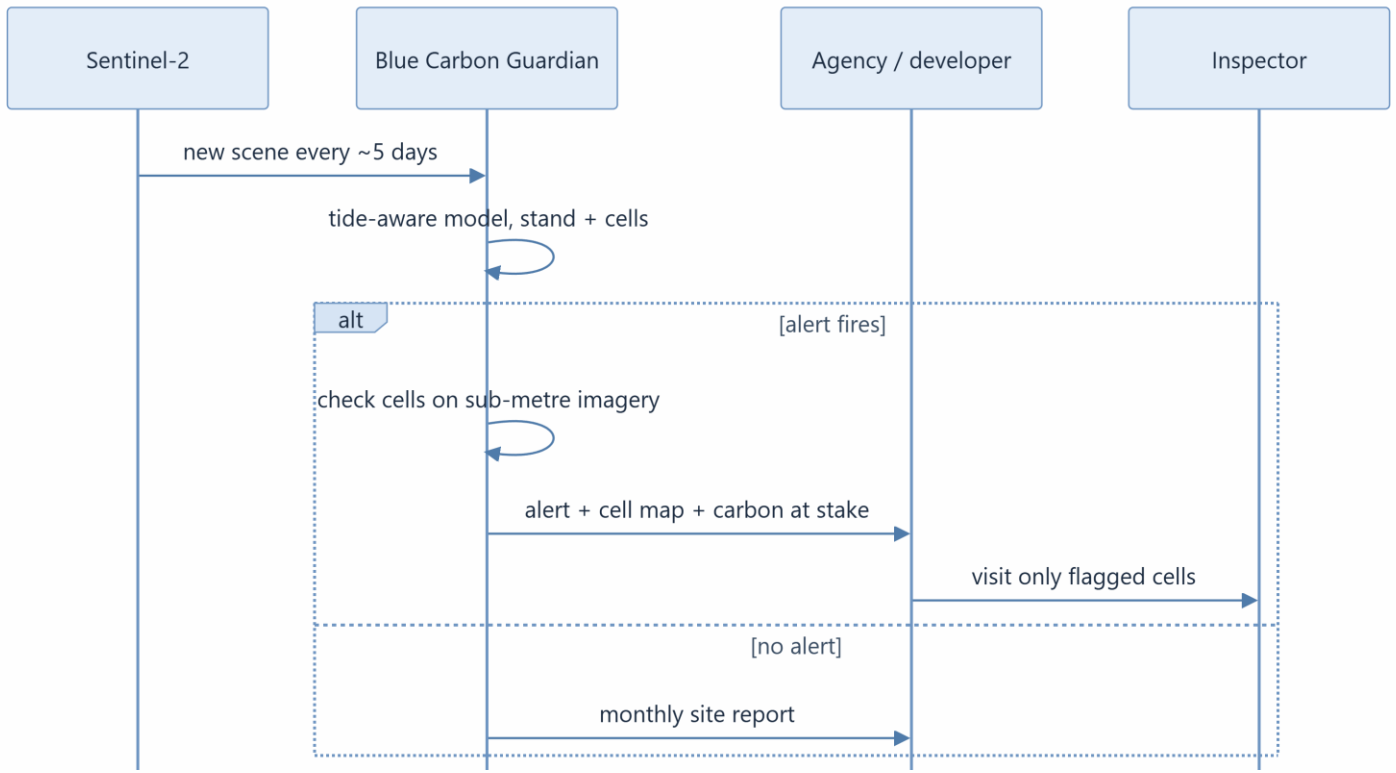
**Every index the monitor uses** (NDVI, red edge, SWIR moisture at 1610 nm, wetness) sits inside 813's ~400–1700 nm range, at 20 m instead of EnMAP's 30 m. **Simulated with EnMAP's 400–1700 nm bands:** the stand-5 loss still reads vegetation → sand and fill (97 % of converted pixels, 0.4 % of intact ones). Not yet on real 813 data.

# From satellite pixels to a decision




Read left to right: Core model · Alert rule · Output to the user · Inputs and checks

# One alert, one message, one targeted visit




### Every ~5 days

a new Sentinel-2 scene is read in place and run through the tide-aware model, per stand and per 200 m cell

### If an alert fires

we check the flagged cells on sub-metre imagery, then send the alert by email or JSON API with the cell map and the carbon at stake

### If nothing changes

the user still gets a monthly site report: condition, trend and carbon stock for every stand

AFTER BAHRAIN SPACE AGENCY FEEDBACK

# Every flag becomes a case

**4 · Urgent** **4 · Monitor** **1 · Closed** date · likely cause · confidence · inspector action · open until a person closes it

### R-05 · Stand 5 — Urgent

| | |
|---|---|
| Open since | 25 Oct 2022 |
| Likely cause | Coastal development: reclamation with finger canals |
| Confidence | **high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree. |
| Inspector action | Inspect the 28 flagged cells (75.3 ha); compare the works boundary with the permit; record the loss for carbon accounting. |

**75.3 ha to inspect**

**2.8–6.7 kt C carbon at stake**

### R-25 · Stand 25 — Monitor

| | |
|---|---|
| Open since | 8 Nov 2024 |
| Likely cause | Unknown: not yet checked on imagery |
| Confidence | **low** · 1 cell(s) below the 2020-21 baseline; one sensor, not yet checked. |
| Inspector action | Check the latest sub-metre image of cell 614 before sending anyone. |

**2.2 ha to inspect**

**up to 0.2 kt C carbon at stake**

### R-03 · Stand 3 — Closed

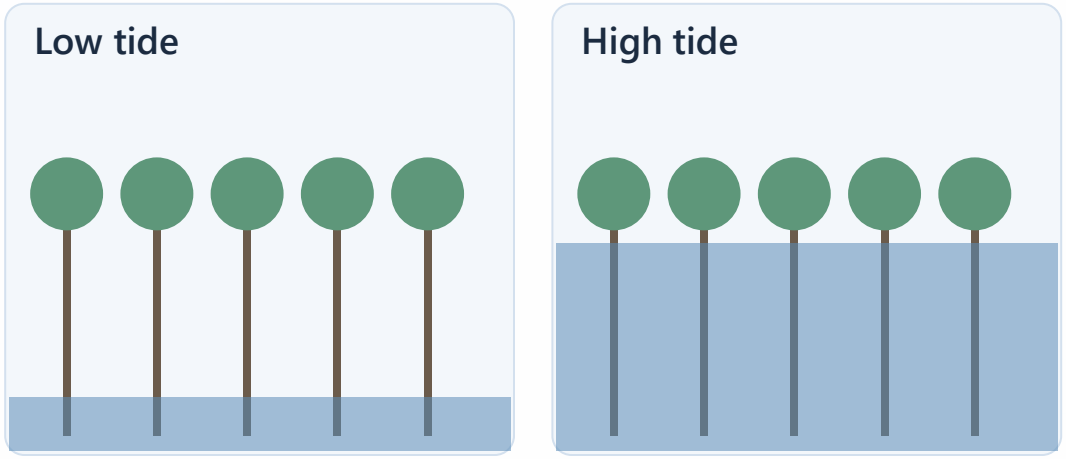
| | |
|---|---|
| Closed on | 26 Oct 2026 |
| Likely cause | Low tide in the image: the canopy was greener than in the previous ten images, so a tide effect, not a loss |
| Confidence | **checked: no loss** · Checked by a person on the image data. |
| Inspector action | None. Closed with the reason recorded. |

Live on the website (Risks section) and as a data feed: api/v1/risks.json

TECHNICAL INNOVATION 1

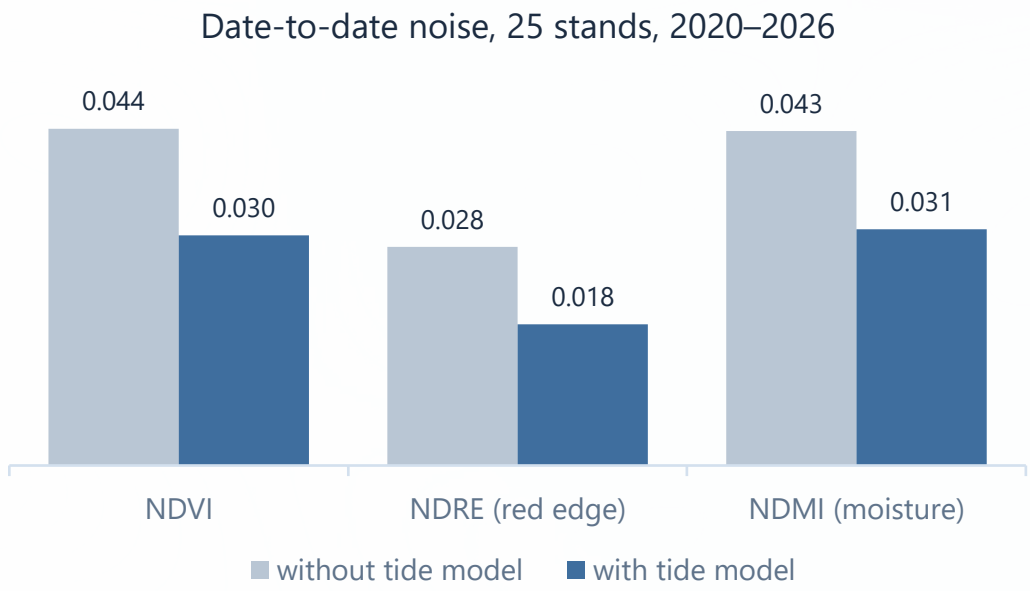
# The tide looks like tree loss — until you correct for it

## WHY SIMPLE MONITORS STRUGGLE ON TIDAL FLATS




NDVI **0.30** NDVI **0.18**

Same trees, same season. Measured on our stands: noise vs wetness r ≈ −0.9.




**−29 to −35 %** less noise on every index: we model trend, season and the tide before flagging anything

**1.61 → 0.96** false alarms per stand-year: NDVI alone vs our tide-aware monitor, same alert rule, same 25 stands (40 % fewer)

# A stand average hides partial loss. 200 m cells do not.




Real stand-5 data, 2022 → 2026: cells turn red while the stand average barely moves.

**Partial loss caught within 60 days (simulated)**

**73 %** stand + cells

**25 %** stand alone

**13 %** by chance

## 8 of 28

cells of stand 9 in severe decline while its average reads “stable”

## 3 of 5

real conversions found only by the cells (stands 12, 18, 6)

### ONE ALERT RULE

stand ≤ −1.5σ OR any cell ≤ −3.5σ

calibrated together: 0.96 false-alarm episodes per stand-year

METHOD IN ONE SLIDE

# How the numbers are made

## TIDE, FROM THE IMAGE ITSELF

**MNDWI = (Green − SWIR) / (Green + SWIR)**
**flooded share = % of pixels with MNDWI > 0**

No tide table: the image shows how much water is under the trees at that exact moment (Sentinel-2 bands B3 and B11).

## EXPECTED VALUE, PER STAND AND CELL

**index ≈ trend + season + tide**
**tide = MNDWI, MNDWI², flooded share**

Robust regression for NDVI, red edge and NDMI, refitted every month on the past 3 years only. The tide terms cut noise by 29–35 %.

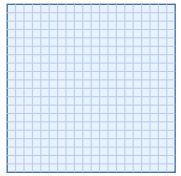
## ALERT

**z = (observed − expected) / robust σ**
**alert: stand ≤ −1.5σ  OR  any cell ≤ −3.5σ**

Median of the last 5 images, so one bad image cannot fire it. Thresholds set on unchanged stands: about 1 false alarm per stand per year.

## WHY 200-METRE CELLS

**200 m = 20 × 20 Sentinel-2 pixels**
**= 400 pixels = 4 ha**



1 cell = 400 px

Enough pixels to average out noise, small enough to send an inspector to 4 ha. 3 of the 5 real losses were found only by cells.



Carbon = area × 108 t C/ha (field plots, Schile et al. 2017). Full equations and sources: README › References and equations.

# How we differ from what exists today

| | Field surveys | One-off satellite maps | Free loss alerts (Global Mangrove Watch) | Blue Carbon Guardian |
|---|---|---|---|---|
| **How often** | campaigns | yearly or less | monthly | **every ~5 days** |
| **Corrects for the tide** | n/a | not usually | not documented | **✓ tide model** |
| **Says where inside a stand** | ✓ | — | — | **✓ 200 m cells** |
| **Published false-alarm rate** | n/a | n/a | accuracy only | **✓ ≈ 1 per stand-year** |
| **Carbon at stake per alert** | — | — | — | **✓ field-based** |
| **Gulf coverage** | ✓ | ✓ | not confirmed | **✓ Abu Dhabi + Tarut Bay** |

**Complementary, not competing:** Global Mangrove Watch flags loss worldwide; we add the tide model, the 200 m cells, a measured false-alarm rate and the carbon at stake — built and tested for the Gulf.

# Five real conversions, confirmed on sub-metre imagery



**Stand 5** reclamation with finger canals



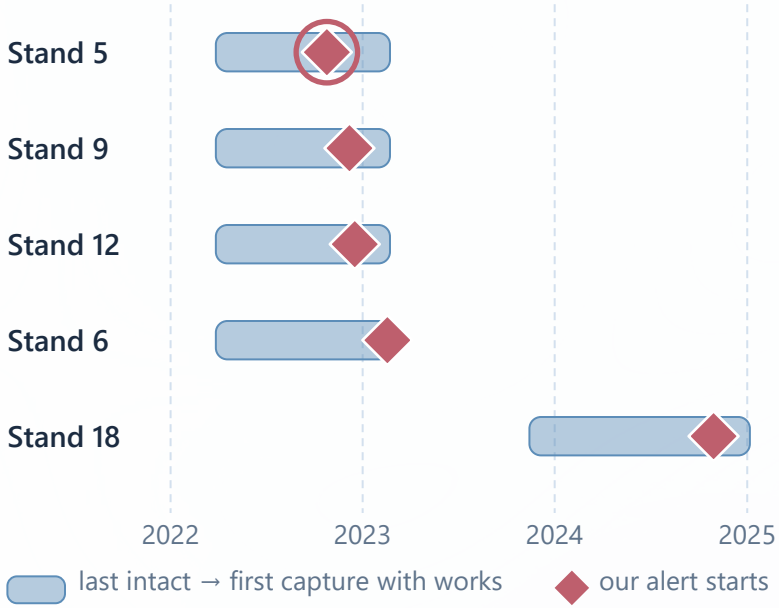
**Stand 9** canal development, south-east




**Stand 12** found only by the cells

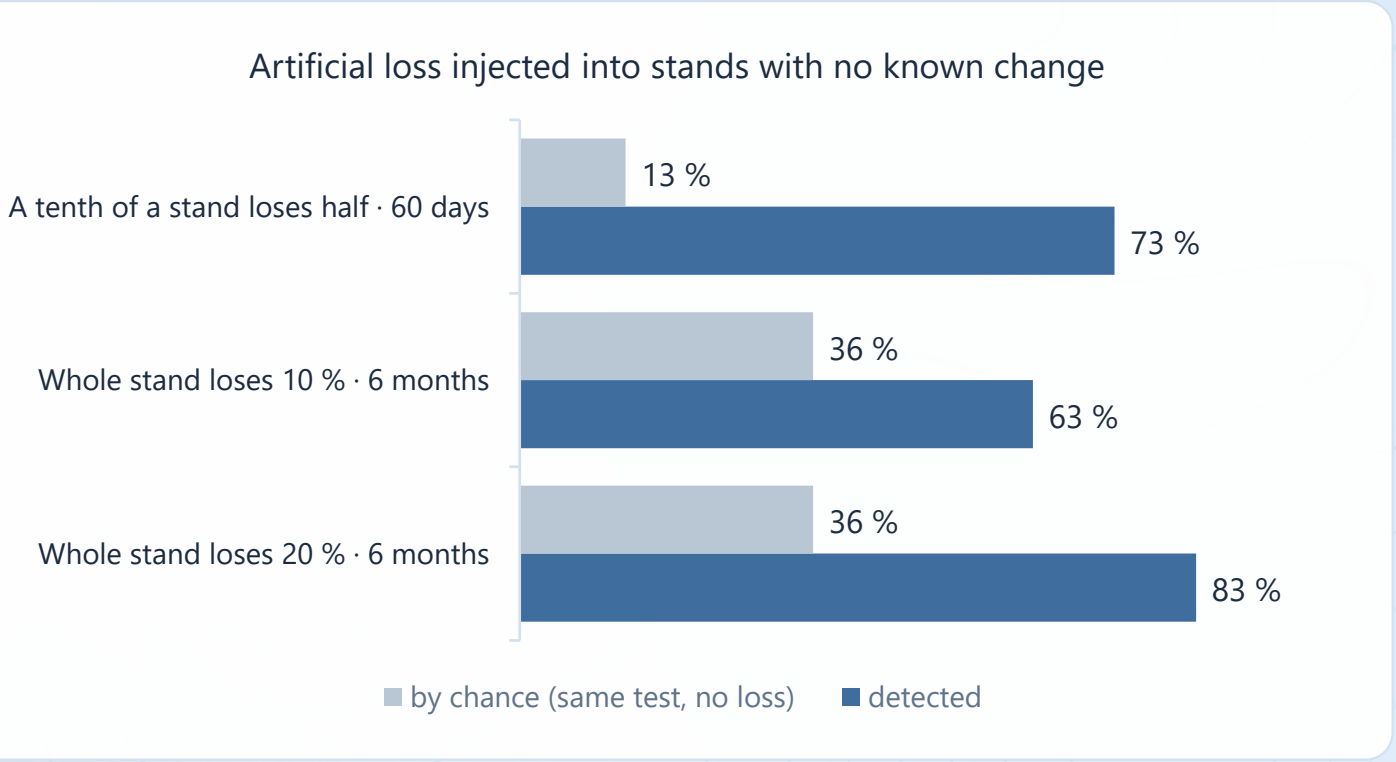



**Stand 18** found only by the cells, 2025




**Every alert began inside the window.** Detection as works happen, not advance warning.

# Every rate next to its chance rate




CALIBRATED · 25 STANDS

0.96

false-alarm episodes per stand-year

UNSEEN · 65 STANDS

0.79

per stand-year, 305 stand-years, incl. Saudi Tarut Bay

ENMAP · SAME DATES

r = 0.84–0.99

vs our Sentinel-2 indices; stand-5 loss 62 ha vs our 26 ha

A detection counts only if a new alert starts after the loss begins. ~160–240 trials per row.

**What we do not claim:** advance warning, field verification, species discrimination, or that hyperspectral classifies better (F1 0.74 vs 0.72).

IMPACT & BUSINESS VIABILITY

# Carbon at stake, and who pays to protect it

## 108 t C/ha

field density near the stands (90 % CI 79–138); 82 % of it is in the soil

## 247 kt C

pilot stock, 2,281 ha (range 180–316) ≈ 907 kt $CO_2e$

## 2.8–6.7 kt C

stock on the 26–62 ha lost at stand 5: what one alert puts on the table

### SERVICE TIERS (PRICES ARE PILOT HYPOTHESES)

**Monitor**

$12k

per site / yr

alert, cell map, condition, carbon, monthly reports

**Evidence**

$25k

per site / yr

+ sub-metre check of every alert, audit pack

**Development watch**

$8k

per project / yr

for coastal developers and EIA consultants

**Cost to serve ≈ $3.3k / site / yr:** compute < $5 (measured), sub-metre checks $500–750, analyst $2k (assumption).

### COMPETITION

Global Mangrove Watch sends free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

### THE HONEST MARKET

Gulf + Red Sea subscriptions ≈ $0.2–0.4 M / yr. The real prize: a **national MRV contract** or a monitoring layer on **gIQ**. SDG 13 · 14 · 15.



Carbon at stake = lost area × field carbon density: 26–62 ha (our Sentinel-2 and EnMAP estimates) × 108 t C/ha (90 % CI 79–138; Schile et al. 2017, Ecol. Appl. 27:859; data Dryad doi 10.15146/R3K59Z) = 2.8–6.7 kt C; × 3.67 = t $CO_2e$. Cost to serve = compute < $5 (measured; Azure D8s v5 $0.384/h) + sub-metre checks $500–750 (SkyWatch $19–29/$km^2$) + analyst $2k (assumption). Prices are pilot hypotheses.

# Reproducible today, built to scale tomorrow

## DOCUMENTATION & EVIDENCE

### Public repository, MIT licence
code, notebook, figures, provenance and licences

### Re-runs end to end in ~24 s
Python 3.12 · Planetary Computer · pinned requirements

### Clean-machine test passed
fresh environment, same numbers, 0 errors

### Every number traceable
each figure comes from a script and a written summary

### Live website + site reports
anyone can open the monitor and check a stand

## DEVELOPMENT POTENTIAL

### Proven now
91 stands · 2 countries · 2 Sentinel-2 tiles · 65 unseen stands at 0.79 false alarms / stand-yr · compute < $5 per site per year

### Incubation
Satellite 813 in the spectral step · deployment on gIQ · a pilot customer + site records · recovery tracking for planting

### Scale
Gulf ≈ 165 km² + Red Sea ≈ 200 km² of mangrove · sub-metre data for small stands (Bahrain) · national MRV, licensing

Areas: Frontiers in Marine Science 2025. Compute measured; prices assumed.

WHY IT MATTERS IN THE REGION

# Big pledges, and a coast that is changing fast



**UAE**

100 M

mangroves by 2030 · MOCCAE, COP26. First GCC monitoring guide: EAD with IUCN, Feb 2026.



**Saudi Arabia**

100 M+

mangroves by 2030 · Saudi Green Initiative. Red Sea Global alone: 50 million by 2030.



**Bahrain**

×4

its mangroves by 2035 · pledged at COP26 (2021). Small stands: they need higher-resolution imagery.



**Arab region**

≈ 365 km²

of mangrove on the Gulf (≈ 165) and Red Sea (≈ 200) coasts; monitoring still "uneven".

5 / 25

pilot stands lost mangrove to coastal development in 2022–25, confirmed on sub-metre imagery (our result).

$25–29

per tonne of $CO_2e$: blue-carbon credit prices 2024–25 (S&P Global Platts). A lost hectare is also lost value.

**Tested: Abu Dhabi + Saudi Tarut Bay · same species across the Gulf (Avicennia marina) · Next: Bahrain, sharper imagery**



Sources: MOCCAE COP26 pledge (2021) · EAD–IUCN GCC mangrove monitoring guide (Feb 2026) · Saudi Green Initiative · Red Sea Global (2023) · Bahrain four-fold by 2035 (GDN, COP26) · Frontiers in Marine Science 2025, doi 10.3389/fmars.2025.1695426 · S&P Global Platts 2024–25

RISK REGISTER · RANKED BY PRIORITY

# What to inspect first

Every flag becomes a case with a date, a likely cause, a confidence level and an action for the inspector. A case stays open until a person checks it and closes it with a reason.

4 · Urgent 3 · Monitor 1 · Closed

## R-05 · Stand 5

Urgent

| | |
|---|---|
| Open since | 25 Oct 2022 |
| Likely cause | Coastal development: reclamation with finger canals |
| Confidence | **high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree. |
| Inspector action | Inspect the 28 flagged cells (75.3 ha); compare the works boundary with the permit; record the loss for carbon accounting. |

75.3 ha to inspect

## R-09 · Stand 9

Urgent

| | |
|---|---|
| Open since | 7 Dec 2022 |
| Likely cause | Coastal development: canals in the south-east |
| Confidence | **high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree. |
| Inspector action | Inspect the 8 flagged cells (21.8 ha); compare the works boundary with the permit; record the loss for carbon accounting. |

21.8 ha to inspect

## R-12 · Stand 12

Urgent

| | |
|---|---|
| Open since | 17 Dec 2022 |
| Likely cause | Coastal development: canal and reclamation at the southern edge |
| Confidence | **high** · Confirmed on dated sub-metre imagery (0.3-0.5 m); stand alert and cells agree. |
| Inspector action | Inspect the 6 flagged cells (13.2 ha); compare the works boundary with the permit; record the loss for carbon accounting. |

13.2 ha to inspect

TEAM BLUE ATHAR · الأثر الأزرق

# Every lost mangrove leaves a trace. We find it from space.

**Overall value: a working product · five real events confirmed · every number reproducible**

| Tide-aware | 5 of 5 | 0.79 | Open |
|---|---|---|---|
| 29–35 % less noise | conversions alerted during the works, 3 only by cells | false alarms / stand-yr on unseen stands | public code, re-runs in ~24 s |

**Next: from PoC to an MVP on GIQ.** In incubation: publish on the GIQ marketplace, pilot with a Gulf agency in Abu Dhabi or Bahrain, add Satellite 813 scenes and recovery tracking.

Scan: live website

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian

BAHRAIN SPACE AGENCY

# Expert Evaluation and Feedback

وكـــالة البحريـن للفضـاء
BAHRAIN SPACE AGENCY

**Specialised assessment and technical feedback**

for research that advances Bahrain's environmental goals

## OFFICIAL STATEMENT

*The Bahrain Space Agency commends Ms. Nahla Nabil for presenting Blue Carbon Guardian, an innovative project using Earth Observation data to monitor regional mangrove ecosystems. The Agency provided specialised assessment and technical feedback, reaffirming its commitment to supporting research and innovation that advance Bahrain's environmental objectives and sustainability goals.*

**— Bahrain Space Agency, official statement**

Meeting and written feedback, 26 Oct 2026 ·
Eng. Roaya Bubshait, Head of Satellite Images & Data Analysis,
with Eng. Aysha Aljawder, Eng. Aysha Almahmeed and Eng. Suad Khalifa

## POTENTIAL ADOPTERS IN BAHRAIN · supporting the national pledge to quadruple mangroves by 2035

*Bahrain's mangrove stands are smaller than Abu Dhabi's: monitoring them needs higher-resolution imagery than open data (10 m) provides.*

**Supreme Council for Environment**
tracking the national mangrove target

**Municipalities & Agriculture**
planting and coastal permits
Ministry of Municipalities Affairs and Agriculture

**Universities**
Earth-observation research
AOU الجامعة العربية المفتوحة Arab Open University البحرين



# Thank you

## شكراً لدولة الإمارات العربية المتحدة

To the United Arab Emirates for leading the region in protecting its mangroves, and for opening space to Arab youth.

**With thanks to:** UAE Space Agency · National Space Academy · Space42 · the mentors and the judges



From Bahrain, with gratitude | Nahla Nabil · Team Blue Athar

REFERENCES

# Where every number comes from

## EXTERNAL SOURCES

**UAE: 100 M mangroves by 2030** Ministry of Climate Change and Environment, COP26 (2021); Abu Dhabi Mangrove Initiative, EAD (2022)

**Saudi Arabia: 100 M+ by 2030** Saudi Green Initiative (sgi.gov.sa); Red Sea Global, 50 M by 2030 (2023)

**Bahrain: ×4 mangroves by 2035** COP26 pledge (2021), GDN Online

**GCC mangrove monitoring guide** EAD with IUCN Mangrove Specialist Group (Gulf News, 24 Feb 2026); Guidelines for Mangrove Restoration in the UAE (EAD, MOCCAE, Emirates Nature–WWF)

**Mangrove areas, "uneven" monitoring** Frontiers in Marine Science (2025), doi 10.3389/fmars.2025.1695426

**108 t C/ha (90 % CI 79–138)** Schile et al. 2017, Ecological Applications 27:859–874, doi 10.1002/eap.1489; field data Dryad doi 10.15146/R3K59Z (CC0)

**$25–29 per t $CO_2e$** S&P Global Platts blue-carbon assessments, 2024–25

**Cost to serve** SkyWatch imagery pricing $19–29/km$^2$; Azure D8s v5 $0.384/h; analyst time = assumption

**Global Mangrove Watch alerts** Bunting et al. 2023, Remote Sensing 15:2050

**Data** Sentinel-2 L2A (ESA Copernicus, via Microsoft Planetary Computer) · EnMAP L2A © DLR 2022, 2025 · ESA WorldCover 2021 (CC BY 4.0) · Esri World Imagery Wayback (Maxar)

## OUR OWN RESULTS · FILE IN THE PUBLIC REPOSITORY

**Tide model: −29 to −35 % noise** analysis/07_summary.md

**40 % fewer false alarms (1.61 → 0.96)** analysis/35_textbook_baseline.md

**73 % vs 13 % by chance** analysis/29_cell_trigger.md, 30_backtest_null.md

**0.96 calibrated · 0.79 on 65 unseen stands** analysis/31_union_alerts.py, 33_scaleout_union_summary.md

**Five real events, alert inside image windows** analysis/34_vhr_summary.md, 26_substand_summary.md

**EnMAP r 0.84–0.99 · 26–62 ha lost** analysis/18_enmap_summary.md

**Satellite 813 simulation (97 % / 0.4 %)** analysis/36_sat813_summary.md

**247 kt C · 2.8–6.7 kt C at stake** analysis/13_carbon_summary.md (area × field density)

**Prices and unit economics** docs/business_plan_draft.md (13 numbered sources)

**Everything re-runs** notebooks/01_blue_carbon_guardian.ipynb · github.com/Nahla-Nabil/blue-carbon-guardian

BACKUP · METHOD

# All the equations behind the numbers

### 1 · Indices (Sentinel-2 bands)

$$NDVI = (B8 - B4) / (B8 + B4)$$
$$NDRE = (B8A - B5) / (B8A + B5)$$
$$NDMI = (B8A - B11) / (B8A + B11)$$

greenness · chlorophyll (red edge) · canopy moisture

### 2 · Tide, from the image

$$MNDWI = (B3 - B11) / (B3 + B11)$$

flooded share f = % of pixels with MNDWI > 0

median over interior pixels of each stand / cell, every image

### 3 · Expected value (per stand, cell, index)

$$index(t) = a + b \cdot t + \text{seasonal (1 and 2 cycles a year)} + e_1 \cdot MNDWI + e_2 \cdot MNDWI^2 + e_3 \cdot f + noise$$

Huber robust regression, refitted monthly on [t − 36 m, t − 3 m]

### 4 · How unusual is today?

$$\sigma = 1.4826 \times \text{median} \mid r - \text{median}(r) \mid$$
$$z = (\text{observed} - \text{expected}) / \sigma$$

state = median of last 5 of mean(z_NDVI, z_NDRE, z_NDMI)

r = residuals of the fit; one bad image cannot fire an alert

### 5 · Alert

ALERT if stand state $\leq -1.5$ OR any cell state $\leq -3.5$

thresholds set on stands with no change → 0.96 false-alarm episodes / stand-year

### 6 · Condition (last 12 months vs 2020–21)

$$\text{change \%} = (\text{now} - \text{base}) / |\text{base}| \times 100$$

severe: NDVI and NDMI $\leq -50$ % · decline: $\leq -15$ % and $\leq -3\sigma$

on the tide-corrected series; now = median of the last 12 months, base = median of 2020–21

### 7 · Carbon

stock = area × 108 t C/ha (90 % CI 79–138)

$CO_2e = C \times 44 / 12$ (= 3.67)

stand 5: 26–62 ha × 108 = 2.8–6.7 kt C

density from 24 field plots near the stands (Schile et al. 2017)

### 8 · Validation

false-alarm rate = alert episodes ÷ stand-years (no change)

detection = % of simulated losses with a NEW alert ≤ 60 days

chance = the same test with no loss → 73 % vs 13 %

simulated loss = a tenth of a stand loses half its canopy

# NDVI alone against our monitor, same alert rule

| Configuration | Indices | Tide model | False alarms per stand-year | Real conversions caught |
|---|---|---|---|---|
| **A · NDVI only** | NDVI | no | 1.61 | 5 of 5 |
| **B** | NDVI | yes | 1.04 | 5 of 5 |
| **C** | NDVI + red edge + SWIR moisture | no | 1.36 | 5 of 5 |
| **D · ours** | NDVI + red edge + SWIR moisture | yes | 0.96 | 5 of 5 |

### 40 % fewer false alarms

20 stands with no known change, 93.9 stand-years: about 16 fewer unnecessary inspections a year for 25 stands.

### The tide model is most of the gain

NDVI alone 1.61 → with tide terms 1.04; the extra indices bring it to 0.96.

### Honest limit

Thresholds calibrated for our monitor; all four catch the five conversions. The difference is the false-alarm cost, not detection.

# Where it breaks — and what we do not claim

### Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.

### 10 m pixels

Patches under ~0.6 ha and young plantings are missed, Bahrain's small stands included.

### Imagery, not site visits

Conversions confirmed on dated sub-metre captures, not yet on permits or field records.

### Simulated losses

Canopy-cover loss, not leaf stress; every rate shown next to its chance rate.

### Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).

### Carbon

Area × field density from four sites (2013–14): a stock with a range, never a sequestration rate.

Rule we kept: no detection rate without its chance rate; no claim of advance warning, field verification or species discrimination.